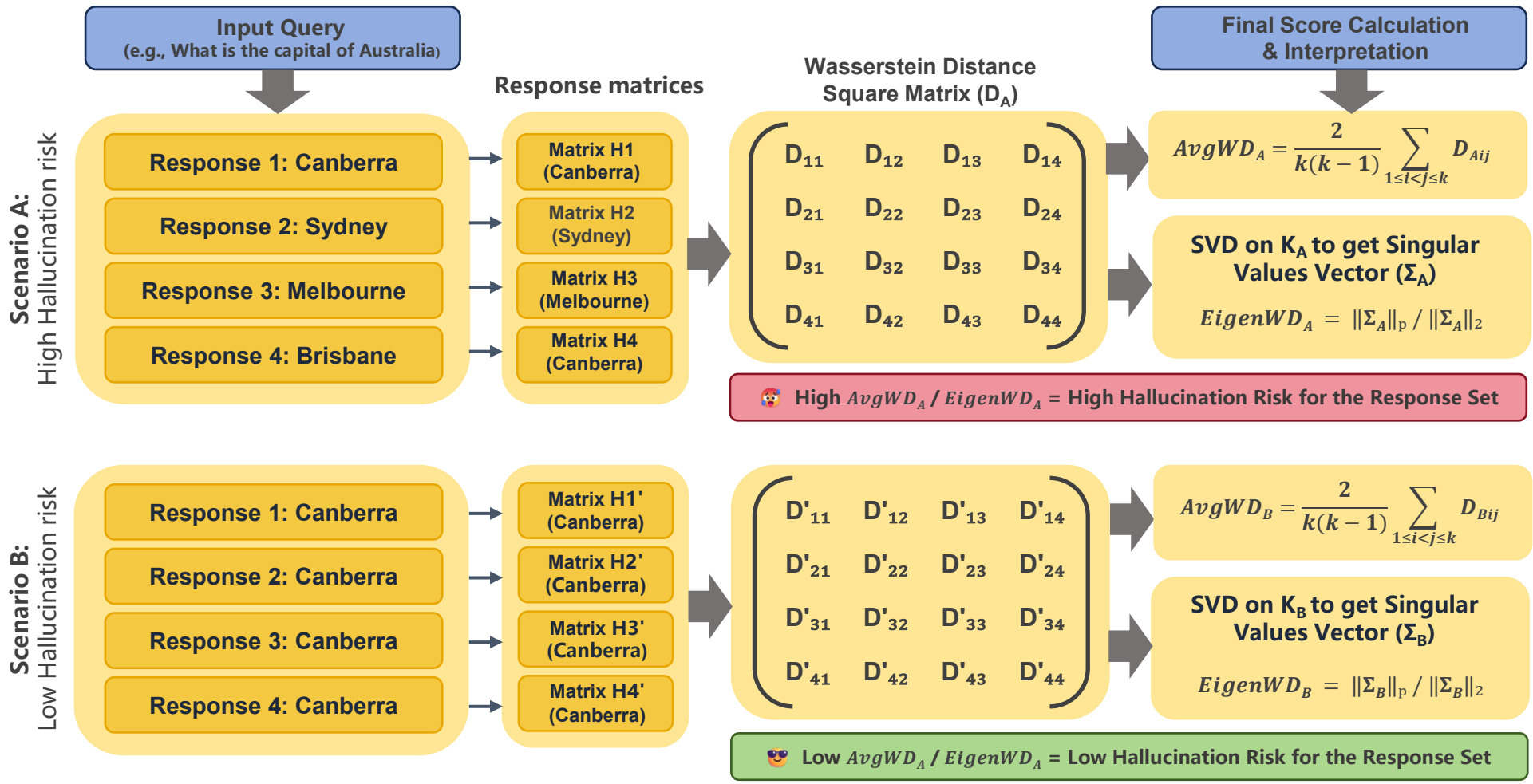
Input Query
(e.g., What is the capital of Australia)
Response matrices
Wasserstein Distance Square Matrix (D_A)
Final Score Calculation & Interpretation
Scenario A: High Hallucination risk
Response 1: Canberra
Response 2: Sydney
Response 3: Melbourne
Response 4: Brisbane
Matrix H1 (Canberra)
Matrix H2 (Sydney)
Matrix H3 (Melbourne)
Matrix H4 (Canberra)
D_{11} D_{12} D_{13} D_{14}
D_{21} D_{22} D_{23} D_{24}
D_{31} D_{32} D_{33} D_{34}
D_{41} D_{42} D_{43} D_{44}
$AvgWD_A = \frac{2}{k(k-1)} \sum_{1 \le i < j \le k} D_{Aij}$
SVD on K_A to get Singular Values Vector (Σ_A)
$EigenWD_A = \|\Sigma_A\|_p / \|\Sigma_A\|_2$
High $AvgWD_A$ / $EigenWD_A$ = High Hallucination Risk for the Response Set
Scenario B: Low Hallucination risk
Response 1: Canberra
Response 2: Canberra
Response 3: Canberra
Response 4: Canberra
Matrix H1' (Canberra)
Matrix H2' (Canberra)
Matrix H3' (Canberra)
Matrix H4' (Canberra)
D'_{11} D'_{12} D'_{13} D'_{14}
D'_{21} D'_{22} D'_{23} D'_{24}
D'_{31} D'_{32} D'_{33} D'_{34}
D'_{41} D'_{42} D'_{43} D'_{44}
$AvgWD_B = \frac{2}{k(k-1)} \sum_{1 \le i < j \le k} D_{Bij}$
SVD on K_B to get Singular Values Vector (Σ_B)
$EigenWD_B = \|\Sigma_B\|_p / \|\Sigma_B\|_2$
Low $AvgWD_A$ / $EigenWD_A$ = Low Hallucination Risk for the Response Set